Individual assignment 5

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(janitor)
library(readxl)
library(snakecase)
library(scales)

# file organization
library(here)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
```

```r
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning data

```r
# creating a new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sampling
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```r
# creating a new object
water_quality_clean <- water_quality |>
  # cleaning column names
  clean_names() |>
  # use a filtering join, only include sites in ncos_sites
  semi_join(ncos_sites,
            by = "site") |>
  # creating new column to join with later data frames
  unite("date_site", date, site,
        remove = TRUE) |>
  # grouping by date/site
  group_by(date_site) |>
  # caluclating median, pH, salinity, and DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungrouping data frame
  ungroup() |>
  # filtering out outliers
  filter(date_site != "2022-08-12_NVBR")
```

```r
# create a new object from taxon_list
taxon_list_clean <- taxon_list |>
  # change names in verbatim_name to snake case
```

```r
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

```r
# create a new object from aq_ins
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filter join to only include sites matching with ncos_sites
  semi_join(ncos_sites,
            by = "site") |>
  # rename column date_on_vial to date
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include FB250/FB 250 samples
  filter(sample_type %in% c("FB250", "FB 250")) |>
  # pivot data frame
  pivot_longer(
    # columns to make longer
    cols = ostracod:annelida_polychaete,
    # new column for the old column names
    names_to = "taxon",
    # new column for the values in each of the old columns
    values_to = "abundance"
  ) |>
  # group by date, site, and taxon
  group_by(date, site, taxon) |>
  # summarize to find average abundance per liter
  summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
  # ungroup
  ungroup() |>
  # combine date and site into date_site, don't remove date and site columns
  unite("date_site", date, site, remove = FALSE) |>
```

```r
  # left join water_quality_clean by date_site column
  left_join(water_quality_clean, by = "date_site") |>
  # left join taxon_list_clean by taxon(aq_ins)/verbatim_name(taxon_list) column
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # mutate to create a column for full names of site codes
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelp Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~ "West Pond",
    site == "NDC" ~ "Devereux Creek"
  ))
```

```r
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

Visualizations from class

```r
# base layer
salinity <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
```

```
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  # different theme
  theme_bw()
```

```
# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(10)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +
  # different theme
  theme_bw()
```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

In part 1d, you should end up with a three panel figure. All three panels should have site on the x-axis, but each panel will have a different y-axis.

In part 1c, you will create one component of the three panel figure. Provide a list of the axes and geometries you will need to create a figure in 1c showing differences between sites in ostracod abundance.

- x-axis: `site_full`
- y-axis: `log_ave_lit`
- geometry to represent average abundance/L: `geom_jitter`
- geometry to represent medians: `stat_summary`

b. Wrangle your data

The `aq_ins_clean` data frame will need some further wrangling before you create the figure you described in part 1a.

In this section, create the data frame you will need to make your figure.

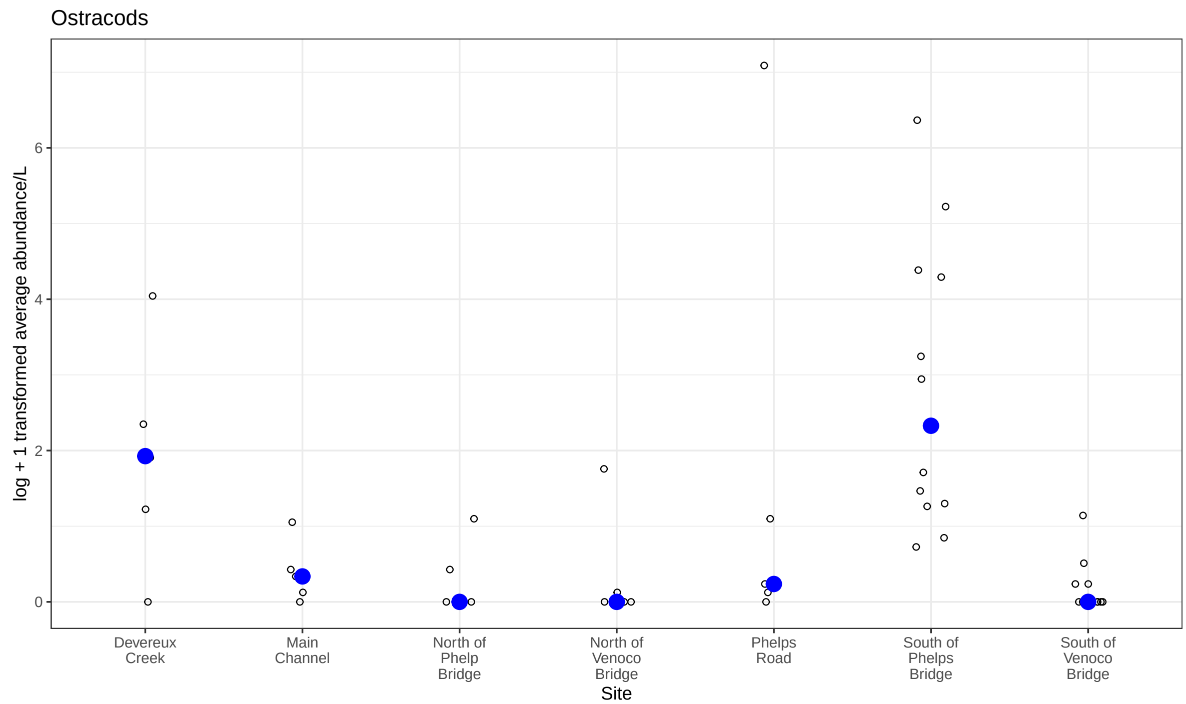
At the end of your code chunk, display 10 random rows from your data frame using `slice_sample()`.

```
ostracods <- aq_ins_clean |>
  filter(taxon == "ostracod") |>
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods, n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2023-03-03_NPB2 2023-03-03 00:00:00 NPB2  ostr~   0.133   7.71    1.98   9.59
 2 2022-05-13_NVBR 2022-05-13 00:00:00 NVBR  ostr~   0       9.72   74.2   11.4
 3 2022-11-19_NDC  2022-11-19 00:00:00 NDC   ostr~   6      NA      NA     NA
 4 2022-11-18_NVBR 2022-11-18 00:00:00 NVBR  ostr~   0      NA      NA     NA
 5 2024-01-28_NPB  2024-01-28 00:00:00 NPB   ostr~   3.33    7.43    3.08   6.6
 6 2022-11-19_NPB1 2022-11-19 00:00:00 NPB1  ostr~   2       7.26    2.47   3.96
 7 2022-01-24_NEC  2022-01-24 00:00:00 NEC   ostr~   0.267  NA      NA     NA
 8 2022-02-25_NEC  2022-02-25 00:00:00 NEC   ostr~   0       8.75   44.1    9.87
 9 2021-08-25_NPB  2021-08-25 00:00:00 NPB   ostr~ 581.      7.52    4.57   4.5
10 2022-08-08_NPB2 2022-08-08 00:00:00 NPB2  ostr~   2       7.68    2.36   2.43
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <chr>, log_ave_lit <dbl>
```

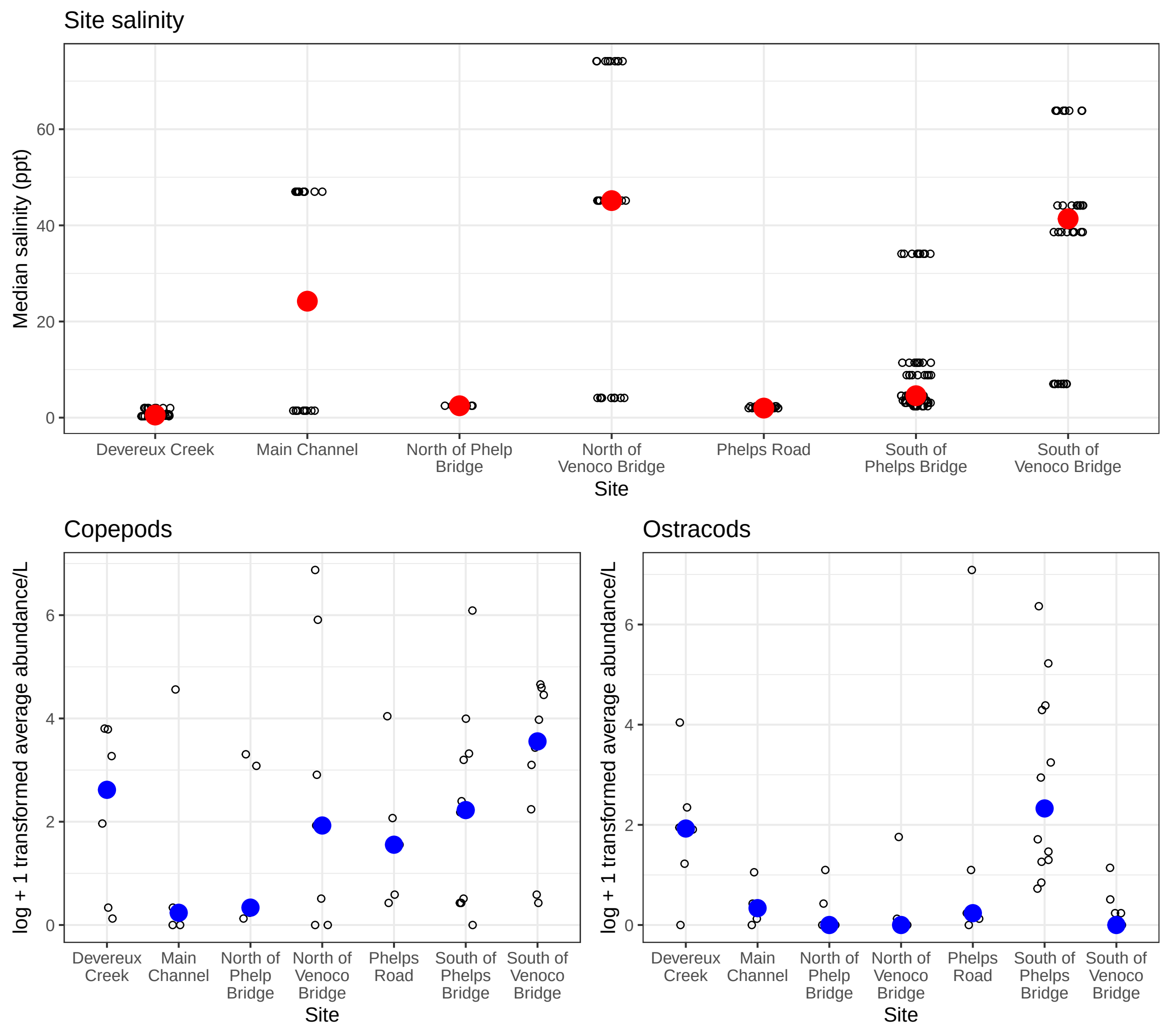
c. Code your figure

```
# base layer
ostracods <- ggplot(data = ostracods,
                    # x-axis: site
                    # y-axis: mean abundance/L
                    mapping = aes(x = site_full,
                                  y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(10)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # different theme
  theme_bw()

# displaying object
ostracods
```



d. Create a multipanel figure

```
salinity / (copepods + ostracods)
```



e. Interpret your figure

In 1-2 sentences each, describe:

- differences between sites in copepod and ostracod abundance
  - On average, there appears to be a higher abundance of copepods across sites compared to ostracods. We can visually tell because of how the blue points representing the central tendency of abundance are generally higher in the y-axis direction in the copepod plot than in the ostracod plot.
- how ostracod abundance may relate (or not) to site-level differences in salinity

- There doesn't appear to be a clear relationship between ostracod abundance and salinity, which we can visually tell because the positions of the median points for ostracod abundance don't appear any higher or lower in response to site salinity (they don't consistently "move" with each other). We might otherwise expect an inverse correlation, but sites like North of Phelps and Phelps Road disrupt this idea.